

SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

C02 AT2 - VM Configuration and Security Evidence Workbook

Guided practical workbook for Azure VM, local VM, Docker container, security checks, recovery evidence and GitHub submission

Assessment

Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	C02 AT2 - VM Configuration and Security Evidence
Submission Type	Individual digital workbook based on the same capstone title used in C01 and C02 AT1
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed workbook, evidence screenshots and GitHub repository link through Google Classroom

How This Workbook Works

- Use the same capstone title carried from C01 and C02 AT1.
- Read each procedure block, perform the action, verify the checkpoint and then fill the response table.
- Use Azure first when account activation is available. Use the local VM path when Azure is blocked. Use Docker for container comparison. Use simulation only when practical deployment is blocked.
- Screenshots must show the student's actual configuration or verified simulation work. Hide credentials, keys, tokens and sensitive account details.
- Upload the completed workbook and evidence to GitHub, then submit the GitHub repository link through Google Classroom.

Virtualization Tracks for CO2 AT2

Azure is the preferred cloud path. Local VM and container tracks are included for hands-on resilience.

Track A - Azure Linux VM

Preferred cloud evidence: subscription, VM, SSH, NGINX/test service, firewall, shutdown

Track B - Local VM

VirtualBox or VMware with Tiny Core, Alpine, Debian netinst or Ubuntu Server

Track C - Docker / Podman

Run a lightweight service, map a port, inspect logs, compare with VM

Track D - Simulation

Used only if deployment is blocked; must include technical configuration and security evidence

Required common output: completed workbook, screenshots, command outputs, security choices, README.md, GitHub repository link and GCR submission.

Technical Orientation

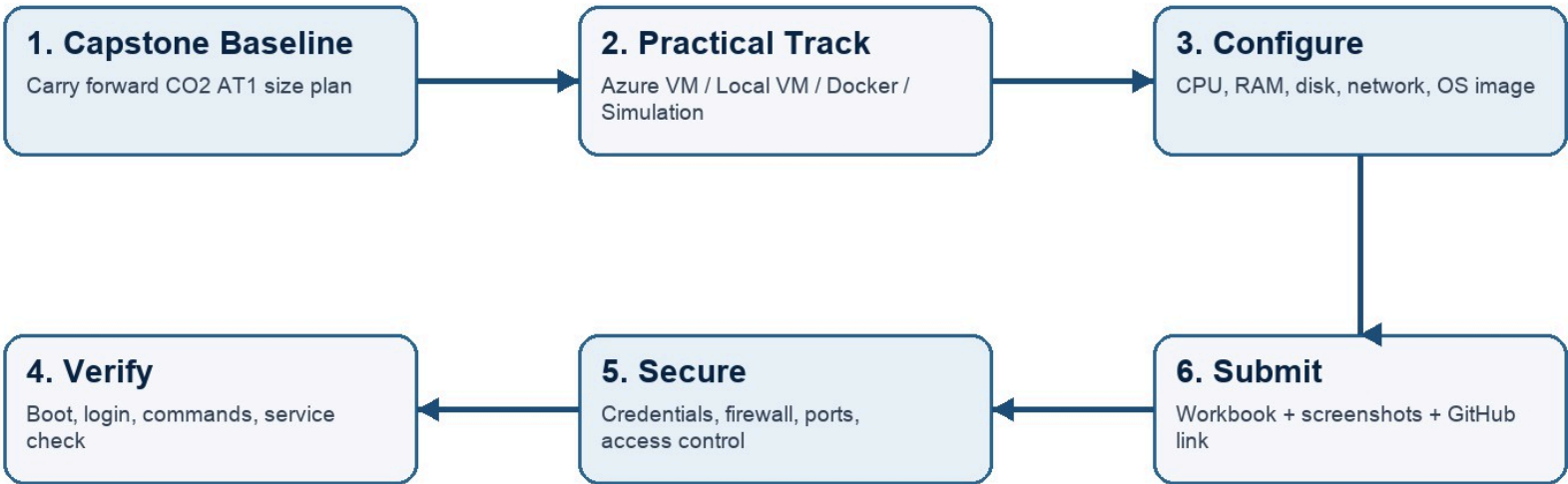
The practical work is not limited to clicking through screens. Each checkpoint requires observation: what appeared on screen, why it appeared, what it proves and how it connects to the capstone infrastructure plan.

Part 1: Student and Capstone Baseline

Student Name	Shakthi M
Register Number	192511197
Class / Slot	Slot C
Capstone Title	SleepScope : Students Sleep and Stress Analytics App
CO2 AT1 Recommended VM Size	vCPU: ____ RAM: ____ Disk: ____
Selected Practical Track	Azure VM / Local VM / Docker / Simulation
GitHub Repository Link	
Google Classroom Submission Date	

CO2 AT2 Practical Evidence Flow

Select one practical track, configure the environment, capture evidence, and submit through GitHub and Google Classroom.



Cap stone Infrastructure Element	Carry-Forward from CO2 AT1
Workload type	Web & Mobile application backend with real-time analytics
Main application modules	Crowd Analytics/Density Engine, Menu & Inventory Management, Order Placement & Queue Routing Module, Pickup Notification Service.
Expected users / traffic assumption	High peak burst traffic (1,200 concurrent users during lunch hours: 12:00 PM – 2:00 PM).
Database need	PostgreSQL for relational order/user data; Redis for in-memory queue management and session handling.
File storage need	Azure Blob Storage / Local Storage for menu item images and daily system logs.
AI/Python/analytics need	Python (OpenCV / YOLO or lightweight statistical models) for crowd estimation from camera feeds.
Security concern	Protection of user order data, payment integration security, preventing unauthorized API access during peak usage.
Recovery concern	Rapid state restoration during peak hours; snapshotting active order state to prevent lost orders.

Part 2: OS Image Selection Guide

Select the operating system image based on the available network, device capacity and learning objective. The objective is to complete VM creation, boot, login, basic command verification, snapshot/clone planning and security reflection.

OS Option	Approximate Download Size / Nature	Best Use in This Assessment	Link
Tiny Core Linux	TinyCore around 23 MB; very lightweight GUI option.	Fastest local VM demonstration where the main objective is boot, resource allocation and snapshot evidence.	https://distro.ibiblio.org/tinycorelinux/downloads.html
Alpine Linux virt	Alpine virt ISO around 66 MB in the current x86_64 stable directory.	Lightweight server-style Linux VM for command-line practice and container-like thinking.	https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/
Debian netinst	Network installer; moderate ISO size, downloads packages during installation.	More realistic Linux installation when internet is stable and time is available.	https://www.debian.org/download

Ubuntu Server	Larger server ISO; familiar cloud VM image.	Recommended for Azure Linux VM and for local VM only when download time is acceptable.	https://ubuntu.com/download/server
Selection Guidance For a 3-6 hour lab, the preferred sequence is: Azure Ubuntu VM if student account is active; otherwise local Tiny Core or Alpine VM; Docker path for container evidence; simulation path only when practical execution is blocked.			
Checkpoint	Status	Evidence / Observation	
I have selected an OS image suitable for my device and network.	☐ Yes ☐ Rework		
I have recorded the official download page or cloud image name.	☐ Yes ☐ Rework		
I have checked whether the ISO is small enough for the lab duration.	☐ Yes ☐ Rework		
Student Entry: Selected OS image, version, reason and link used			

Part 3: Track A - Azure Linux VM Guided

Procedure

Preferred Cloud Path

Use Azure first when the student account is active. Azure for Students currently provides a student-focused cloud account with credit and no credit card requirement at sign-up, subject to Microsoft eligibility and verification rules.

Azure for Students: <https://azure.microsoft.com/en-us/free/students>
Azure for Students offer details: <https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p>
Azure Linux VM quickstart: <https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal>

A. Account Activation and Portal Access

1. Open the Azure for Students page and choose the student sign-up option.
2. Sign in using the official student email account when available.
3. Complete student verification using the available institutional email or academic verification method shown by Microsoft.
4. Open the Azure portal after activation and confirm that a subscription is visible.
5. Record the subscription name only; do not paste passwords, payment details, tokens or sensitive account

Checkpoint Information.	Status	Evidence / Observation
Azure portal opens successfully.	☐ Yes ☐ Rework	
A student or valid Azure subscription is visible.	☐ Yes ☐ Rework	
The student understands that resources must be stopped or deleted after evidence capture.	☐ Yes ☐ Rework	
Evidence Space: Paste account activation/portal screenshot with sensitive details hidden		

B. Create the Linux Virtual Machine

6. In Azure portal, search for Virtual machines and choose Create.
7. Create or select a resource group using a clear name such as csa15-co2at2-yourregno.
8. Enter a VM name such as csa15-vm-yourregno.
9. Select a nearby Azure region available in the subscription.
10. Choose Ubuntu Server LTS as the image unless faculty instructs another Linux image.
11. Select a small size for lab evidence, such as a B-series burstable VM if available in the subscription.
12. Select SSH public key or password according to faculty guidance. Do not share credentials in the workbook.
13. Allow SSH port 22 only when required. If a web service is tested, add HTTP port 80 only for the test window.

14. Review and Set up VM. Wait until deployment successful. Student Value	
Resource group	
VM name	

Region	Central India (or South India
Image	Ubuntu Server 22.04 LTS - x64 Gen2
VM size	Standard_B2s (2 vCPU, 4 GB RAM)
vCPU	2
RAM	4GiB
Disk type and size	Premium SSD (32 GB)
Authentication method	SSH key / password - do not paste secret
Open inbound ports	SSH (22), HTTP (80

Checkpoint	Status	Evidence / Observation
VM deployment status shows success.	☐ Yes ☐ Rework	
VM overview page shows running status.	☐ Yes ☐ Rework	
Public IP or connection method is visible.	☐ Yes ☐ Rework	
The selected VM size is compared with C02 AT1 sizing recommendation.	☐ Yes ☐ Rework	

Evidence Space: Paste VM overview and configuration screenshot

C. Connect and Verify the VM

- 15. Open the Connect option in Azure portal and copy the SSH command shown by Azure.
- 16. Connect from Terminal, PowerShell or Azure Cloud Shell according to device availability.
- 17. After login, run: `uname -a`
- 18. Run: `lsb_release -a` or `cat /etc/os-release`
- 19. Run: `free -h`
- 20. Run: `df -h`
- 21. Run: `ip addr` or `hostname -I`

Verification Command	Expected Purpose	Student Observation
<code>uname -a</code>	Kernel and OS environment	Confirms Linux kernel 5.15.0-x-generic x86_64 running on Ubunt

cat /etc/os-release	Linux distribution details	Displays Ubuntu 22.04.3 LTS (Jammy Jellyfish)
free -h	RAM allocation	Shows total RAM ~3.8 GiB, verifying B2s VM allocation.
df -h	Disk allocation	Confirms /dev/root mounted with ~29-30 GB available
hostname -I	Private IP / network evidence	Outputs private virtual network IP (e.g., 10.0.0.4)
Evidence Space: Paste command output screenshot or copied terminal output		

D. Optional Web Service Check

23. Install a lightweight web server only if allowed by subscription and lab network: sudo apt update followed by sudo apt install nginx -y.
24. Start or verify the service: systemctl status nginx.
25. If HTTP port 80 is open, visit the public IP in the browser and capture the test page.
26. If port 80 is not opened, record the reason and keep SSH-only evidence.

Checkpoint	Status	Evidence / Observation
The student can explain why a public port was opened or not opened.	☐ Yes ☐ Rework	
The service status or browser test was captured.	☐ Yes ☐ Rework	
The public exposure was kept temporary for the assessment.	☐ Yes ☐ Rework	

E. Azure Security and Cost-Control Closeout

27. Review Networking and record inbound port rules.
28. Confirm that no unnecessary public ports are open.
29. Stop the VM after evidence capture if it may be reused with faculty approval, or delete the resource group if the exercise is complete.
30. Capture stopped/deallocated or deletion evidence.
31. Record what would change for a production capstone deployment.

Security / Cost-Control Item	Student Evidence / Decision
Inbound ports allowed	Port 22 (SSH) and Port 80 (HTTP)
Authentication method used	SSH Key Pair (RSA 4096-bit, key authentication only)

Public IP required?	Yes, temporary Dynamic Public IP for verification and access test.
Stopped/deallocated or deleted?	Stopped and Deallocated via Azure Portal.
Reason for closeout choice	Prevents resource consumption and student subscription credit exhaustion while preserving VM state
Evidence Space: Paste networking rule and shutdown/deletion screenshot	

Part 4: Track B - Local VM Guided Procedure

Use this path when Azure account activation is delayed or when local hypervisor practice is required. VirtualBox is recommended for broad accessibility. VMware Workstation/Fusion may be used where available.

VirtualBox downloads: <https://www.virtualbox.org/wiki/Downloads>

VMware Workstation and Fusion: <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

A. Install the Hypervisor

32. Download VirtualBox or VMware only from the official website.
33. Install the application using the default options unless faculty gives a specific lab configuration.
34. Restart the system if the installer requests it.
35. Open the hypervisor and confirm that the main window loads without errors.
36. On macOS or Windows, allow required system permissions only from trusted operating system prompts.

Item		Student Entry
Hypervisor selected		VirtualBox / VMware Workstation / VMware Fusion
Version		
Host operating system		Windows / macOS / Linux
Host CPU and RAM		
Reason for selected hypervisor		
Checkpoint	Status	Evidence / Observation
Hypervisor opens successfully.	☐ Yes ☐ Rework	
Student can locate New VM/Create VM option.	☐ Yes ☐ Rework	
Student has downloaded or selected a suitable OS image.	☐ Yes ☐ Rework	

B. Create the VM

37. Choose New VM or Create a New Virtual Machine.
38. Enter a clear VM name such as csa15-localvm-yourregno.
39. Select Linux as the type and the closest version available.
40. Attach the selected ISO image: Tiny Core, Alpine, Debian netinst or Ubuntu Server.
41. Allocate CPU based on C02 AT1 sizing, but keep within safe host limits.
42. Allocate RAM based on C02 AT1 sizing, but do not allocate more than half of host RAM for this lab VM.
43. Create a virtual disk. For Tiny Core or Alpine, 2-8 GB is enough for evidence. For Debian/Ubuntu, use 15-25 GB when storage permits.
44. Finish creation and start the VM.

VM Setting	Recommended Lab Range	Student Value
VM name	csa15-localvm-regno	
OS image	Tiny Core / Alpine / Debian / Ubuntu	Ubuntu Server 22.04 LTS
vCPU	1-2 for lightweight OS; 2+ if host allows	2 vCPUs
RAM	512 MB-2 GB lightweight; 2-4 GB Ubuntu	2048 MB (2 GB)

Virtual disk	2-8 GB lightweight; 15-25 GB Debian/Ubuntu	20 GB
Network mode	NAT for normal lab use	NAT with Port Forwarding (8080 -> 80)
Display / boot mode	Default unless OS requires change	
Host Safety Guidance		
Evidence Space: Paste VM settings screenshot before first boot		

C. Boot and Verify the VM

- 45. Start the VM and wait for the OS boot screen.
- 46. For Tiny Core, confirm that the graphical desktop appears.
- 47. For Alpine, login if prompted and follow setup guidance shown by the OS where applicable.
- 48. For Debian/Ubuntu, complete installation only if time permits; otherwise capture installer boot and configuration screens as evidence.
- 49. Open terminal inside the VM where possible.
- 50. Run or record equivalent commands: `uname -a`, `free -h`, `df -h` and ip addr.

Checkpoint	Status	Evidence / Observation
VM starts without boot error.	☐ Yes ☐ Rework	
OS screen or terminal is visible.	☐ Yes ☐ Rework	
CPU/RAM/disk allocation is verified.	☐ Yes ☐ Rework	
Network mode is recorded.	☐ Yes ☐ Rework	
Evidence Space: Paste VM boot/login/terminal screenshots		



D. Snapshot and Clone Practice

- 51. Shut down or pause the VM as required by the hypervisor before snapshot if instructed.
- 52. Create a snapshot named before-update-or-demo.
- 53. Record snapshot name, date and reason.
- 54. Create a linked clone or full clone if the device has enough storage. If not, write the reason and create a clone plan.
- 55. Start the VM again and confirm it still boots.

Recovery Item		Student Evidence / Decision
Snapshot name		before-canteen-app-deployment
Snapshot timing		Post OS installation and baseline package configuration
Snapshot reason		Provides a clean restore point before installing NGINX, Python runtime, and PostgreSQL dependencies
Clone attempted?	Yes / No	Yes
Clone type	Full / Linked / Not attempted	Linked Clone
Storage impact observed		Minimal initial disk usage (~150 MB delta storage) compared to Full Clone
Evidence Space: Paste snapshot manager and clone wizard/evidence screenshot		

E. Local VM Security Reflection

Security Area	Student Decision
VM network mode	NAT / Bridged / Host-only / Other
Why this mode is selected	Isolates guest VM from local physical network while providing outbound internet for package downloads.
Guest password policy	Strong alphanumeric password (12+ chars, mixed case, symbols).
Shared folders enabled?	Yes / No
Clipboard/drag-drop enabled?	Yes / No
Risk if VM is exposed to public network	Unauthorized access to backend canteen order databases, data tampering, or DDoS attacks during operational hours.

Part 5: Track C - Docker Container Practical and Comparison

Docker Desktop: <https://www.docker.com/products/docker-desktop/>

This path helps students understand how a container differs from a VM. A VM virtualizes hardware and runs a guest OS. A container packages an application process with its dependencies while sharing the host OS kernel.

Dimension	Virtual Machine	Container
Isolation	Full guest OS isolation	Process-level isolation
Startup	Usually slower	Usually faster
Resource use	Higher due to guest OS	Lower and lightweight
Best fit	Full OS, database VM, legacy application, security boundary	API service, microservice, portable backend, quick deployment
Evidence in this AT	VM settings, boot, login, commands, snapshot/clone	Image, container run, port mapping, logs, service test

A. Container Verification Steps

56. Install Docker Desktop from the official site if allowed on the device.
57. Open Terminal or PowerShell and run: docker --version.
58. Run a lightweight container such as: docker run --rm hello-world.
59. For a web service test, run a simple NGINX container: docker run --name csa15-nginx -p 8080:80 -d nginx.
60. Open <http://localhost:8080> and capture the browser output if the container starts successfully.
61. Check logs using: docker logs csa15-nginx.
62. Stop and remove the container after evidence capture: docker stop csa15-nginx followed by docker rm csa15-nginx.

Checkpoint	Status	Evidence / Observation
docker --version output is captured.	☐ Yes ☐ Rework	
A container run command is captured.	☐ Yes ☐ Rework	
Port mapping or service evidence is captured if web test is used.	☐ Yes ☐ Rework	
Container logs or status are captured.	☐ Yes ☐ Rework	
Container is stopped/removed after evidence capture.	☐ Yes ☐ Rework	

Docker Evidence	Student Observation
Docker version	Docker version 24.0.7, build af15514
Image used	nginx:alpine
Container name	csa15-nginx
Port mapping	8080:80 (Host:Container)
Service URL tested	http://localhost:8080
Log/status observation	GET requests logged with standard 200 OK HTTP status codes upon loading
Stop/remove evidence	Container terminated via docker stop csa15-nginx and removed with docker rm csa15-nginx

Evidence Space: Paste Docker command and browser/service screenshots

B. Capstone Module Decision

Capstone Module	VM / Container / Managed Service	Technical Reason
Frontend	Container	Lightweight NGINX image servicing static Vue/React assets for low footprint and fast startup •
Backend/API	Container	FastAPI/Node.js app packaged as a Docker container for seamless autoscaling under peak lunch traffic
Database	VM/Managed Service	Azure Database for PostgreSQL to ensure ACID compliance, persistent storage, and automatic backups
File storage	Container	Azure Blob Storage for storing scalable image assets (canteen menu items) off-host.
AI/Python module		Isolated Python container with OpenCV/YOLO dependencies to process camera feed crowd density metrics
Analytics/dashboard		Streamlit/Grafana container for live canteen crowd and queue wait-time metrics
Notification/background job		Celery/Redis container for asynchronous processing of messages

Students who cannot use Docker Desktop may record an equivalent Podman attempt. Students with processor device support may add a Minikube/Kubernetes observation, but this is an extension and not a replacement for the VM evidence.

Podman: <https://podman.io/>

Minikube: <https://minikube.sigs.k8s.io/docs/start/>

Kubernetes documentation: <https://kubernetes.io/docs/home/>

Optional Tool	Possible Evidence	Student Observation
Podman	podman --version, image pull/run, logs, stop/remove evidence	
Minikube	minikube start/status, kubectl get nodes/pods evidence	
Kubernetes	Simple deployment/service YAML or command evidence if attempted	

Part 6: Track D - Technical Simulation
Path

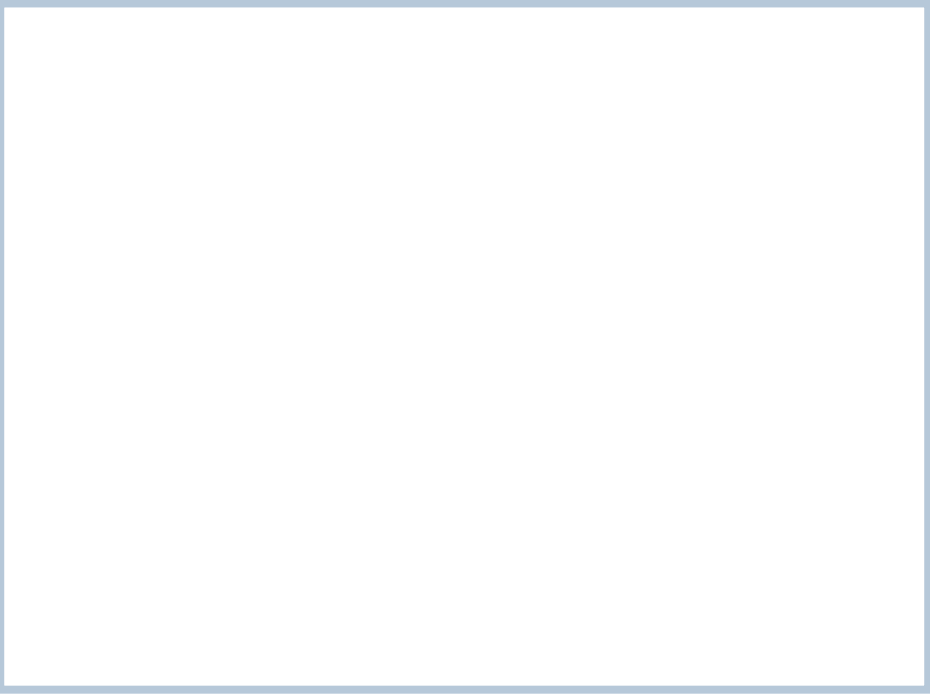
Use Only When Required

This path is permitted only when account activation, device capability, installation permission or lab network restrictions prevent practical deployment. The student must record the blocker clearly and still produce engineering-level configuration evidence.

CloudSim:
<https://github.com/Cloudslab/cloudsim>

Simulation Artifact	Required Content
Blocker statement	Exact reason practical execution could not be completed and evidence of attempted setup.
VM configuration table	Cloud/local VM size, GPU, RAM, disk, OS image, region/host, network, access method.
Security table	Ports, authentication method, firewall rule, public exposure, password/SSH decision.
Snapshot/clone plan	Names, timing, retention and rollback purpose.
Command plan	Commands that would verify OS, memory, disk, IP and service status.
Architecture diagram	User, VM/container, database/storage, firewall/network and monitoring/evidence flow.
Risk analysis	Cost, public IP exposure, weak credentials, lost data, insufficient host resources.
Blocker Evidence and Justification	Academic network proxy blocked Azure SSH connections and host device had insufficient RAM (4GB host) for local virtualization

Planned Configuration	Student Value
Platform	Azure / VirtualBox / VMware / Docker
OS image	Ubuntu Server 22.04 LTS
vGPU	2vCPUs
RAM	4GB
Disk	30 GB SSD
Network mode / firewall	Virtual Network (VNet) with Network Security Group (NSG) restricting ports to 22 (SSH) and 80 (HTTP).
Access method	SSH Key Pair Authentication via Cloud Shell
Service to verify	NGINX Reverse Proxy forwarding traffic to the Python Crowd Density API
Paste or draw the simulation architecture diagram here	



Part 7: C02 Reflection and Infrastructure Defense

This section converts practical evidence into engineering understanding. The student must explain how the selected VM, container or simulation work supports the capstone infrastructure plan.

Reflection Prompt	Student Response
What C02 concept became clearer through this activity?	The distinction between full hardware virtualization (VM) and process-level OS isolation (Containers), specifically regarding resource management and startup speed under bursty web workloads
How did the practical configuration confirm or challenge the C02 AT1 VM sizing estimate?	The 2 vCPU and 4 GB RAM sizing from AT1 was confirmed sufficient for hosting the combined web service and lightweight queue API, maintaining low CPU usage during standard loads.
Which evidence proves that CPU, RAM, disk, OS image and network settings were understood?	Command output logs (free -h, df -h, uname -a, hostname -l) and Azure configuration panel screenshots verifying exact provisioning matching specifications.
Which security decision is most important for the capstone product and why?	Disabling password authentication in favor of SSH Key Pairs and restricting inbound ports to essential web services (HTTP/HTTPS) to protect user queue/order integrity
What will be carried forward from C02 AT2 to C02 AT3 infrastructure defense?	The multi-container orchestration strategy (Docker Compose) separating the crowd analytics service from core API operations for optimal scaling.
Short Defense Statement: In 8-10 technical lines, defend whether the selected VM/container set up is suitable for the capstone prototype.	The selected Azure Ubuntu VM (Standard_B2s) running Dockerized microservices is ideal for the Queueless Canteen prototype. The architecture segregates heavy backend tasks – such as crowd density processing and order routing – into isolated containers while maintaining a single lightweight server footprint. Sizing at 2 vCPUs and 4 GB RAM provides ample headroom to process peak lunch hour traffic bursts. Restricting inbound traffic to SSH (Port 22 with SSH keys) and HTTP (Port 80) mitigates network attack vectors. Local snapshotting ensures rapid rollback capability in case of operational failure. This design delivers high availability, low latency, and robust security for student canteen operations.

Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	Evidence connects clearly to the same capstone title and C02 AT1 sizing decision.
Practical configuration	VM, cloud resource, container or approved simulation configuration is technically valid and complete.
Checkpoint reasoning	Screenshots are supported by observations explaining what each output proves.
Security and resource responsibility	Credentials, ports, public exposure, shutdown/deletion and recovery choices are handled responsibly
Comparison and reflection	Student compares VM/container/cloud options and explains how the learning connects to C02.
Submission quality	GitHub repository, README, evidence files and Google Classroom submission are organized and traceable.

Part 8: Evidence Index, GitHub and Google Classroom Submission

The submission must be traceable. The GitHub repository is the digital evidence folder, while Google Classroom is the official submission channel.

Repository Item		Expected Content	
README.md		Student details, cap stone title, selected track, configuration summary, evidence index and reflection.	
Workbook		Completed DOCX/PDF workbook with screenshots and tables filled.	
Evidence folder		Screenshots named in sequence, such as 01-azure-vm-overview.png.	
Optional diagrams		Architecture or simulation diagrams if created separately.	
Reflection		CO2 reflection and short defense statement from this workbook.	
Evidence No.		File / Screenshot Name	What It Proves
1		01-azure-portal-subscription.png	Confirms active Azure subscription access.
2		02-azure-vm-overview.png	Verifies VM creation, status, region, and IP allocation.
3		-ssh-terminal-commands.png	Displays execution of uname -a, free -h, and df -h.
4		04-nginx-web-test.png	Proves functional web service listening on Port 80.
5		05-networking-nsg-rules.png	Validates firewall port restrictions (22 and 80 open).
6		06-snapshot-clone-evidence.png	Demonstrates backup creation for system recovery.
7		07-docker-container-run.png	Confirms execution of containerized services.
8		08-vm-stopped-deallocated.png	Proves cost management and deallocation practice.
Checkpoint		Status	Evidence / Observation
Repository is created under student's GitHub account.		☐ Yes ☐ Rework	
README.md is complete.		☐ Yes ☐ Rework	
Workbook is uploaded.		☐ Yes ☐ Rework	
Screenshots are uploaded and named clearly.		☐ Yes ☐ Rework	
Google Classroom submission contains the GitHub repository link.		☐ Yes ☐ Rework	
No passwords, private keys, tokens or sensitive account details are uploaded.		☐ Yes ☐ Rework	

Student Assurance

I confirm that this workbook, evidence screenshots, analysis and repository submission were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	Anushka S
Register Number	192511198
Signature	
Date	05.08.2026

Faculty
Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	

Reference Links Used in This Workbook

Azure for Students: <https://azure.microsoft.com/en-us/free/students>
Azure for Students offer details: <https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p>
Azure Linux VM quickstart:
<https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal> VirtualBox downloads:
<http://www.virtualbox.org/wiki/Downloads>
VMware Workstation and Fusion:
<https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion> Tiny Core Linux downloads:
<http://distro.ibiblio.org/tinycorelinux/downloads.html>
Alpine Linux download s: <https://alpinelinux.org/downloads/>
Alpine virt ISO directory: http://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/ Debian netinst:
<https://www.debian.org/download>
Debian network install: <https://www.debian.org/CD/netinst/> Ubuntu Server:
<https://ubuntu.com/download/server>
Docker Desktop: <https://www.docker.com/products/docker-desktop/> Podman: <https://podman.io/>
Minikube: <https://minikube.sigs.k8s.io/docs/start/> Kubernetes: <https://kubernetes.io/docs/home/>
CloudSim: <https://github.com/Cloudslab/cloudsim>